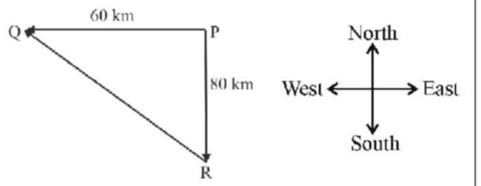
(a) 140 km (b) 130 km
(c) 110 km (d) 100 km

Ans : (d)

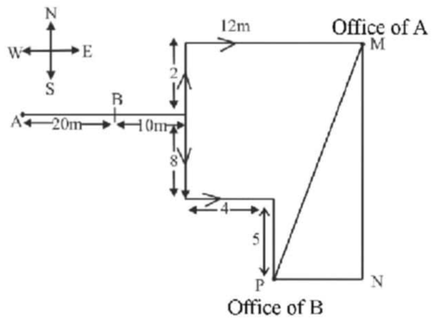


In Δ PQR,
$QR^2 = QP^2 + PR^2$
$QR^2 = (60)^2 + (80)^2$
$QR^2 = 3600 + 6400$
$QR^2 = 10,000$
$QR = 100Km$

43. 'A' started from his house and walked 20m towards East, where his friend 'B' joined him. They together walked 10m in the same direction. Then 'A' turned left while 'B' turned right and travelled 2m and 8m respectively. Again 'B' turned left to travel 4m followed by 5m to his right to reach his office. 'A' turned right and travelled 12m to reach his office. What is the shortest distance between the two offices?

(a) 15 m (b) 17 m
(c) 19 m (d) 20 m

Ans : (b)



From figure,
PN = (12 – 4)
= 8m.
MN = (2 + 8 + 5)
= 15m
By pythagorus theorem, In Δ PNM
$PM^2 = MN^2 + PN^2$
$PM^2 = (15)^2 + (8)^2$
$PM^2 = 225 + 64$
$PP^2 = 289$
$PM^2 = 17$

GENERAL MENTAL ABILITY

Direction for the following 3 (three) items:
Read the following information and answer the three items that follow:
Six students A,B,C,D,E and F appeared in several tests. Either C or F scores the highest. Whenever C scores the highest, then E scores the least. Whenever F scores the highest, B scores the least.
In all the tests they got different marks; D scores higher than A, but they are close competitors; A scores higher than B; C scores higher than A.

44. If F stands second in the ranking, then the position of B is-

(a) Third (b) Fourth
(c) Fifth (d) Sixth

Ans : (c) According to question,
Conditions will be as follows:-
स्थितियाँ निम्न होगी-

Condition-I	Condition-II
1→ C	1→ F
2	2
3	3
4	4
5	5
6 → E	6 → B

Condition III:-

C
D
A
B

If F stands second in the ranking then the position of B-
∴ 1→ C
2→ F
3 → D
4 → A
5 → B
6 → E
Hence, The position of B is fifth.

45. If B scroes the least, the rank of C will be-

(a) Second
(b) Third
(c) Fourth
(d) Second or third

Ans : (d) If B scores the least, for the position of C, from conditions II[nd] & III[rd]

Ranking	Condition-I	Condition-II
1. _____	F_____	F
2. _____	C_____	__
3. _____	D_____	C
4. _____	A _____	D
5. _____	__ _____	A
6. _____	B _____	B

Hence,
The position of c will be either II[nd] or III[rd]

46. If E is ranked third, then which one of the following is correct?

(a) E gets more marks than C
(b) C gets more marks than E
(c) A is ranked fourth
(d) D is ranked fifth

Ans : (b)
If E is ranked third, from condition II[nd] & III[rd]
1→F
2→C
3→E
4→D
5→A
6→B
Hence, C gets more marks than E

Directions for the following 2 (two) items-
Read the following statements S1 and S2 and answer the two items that follow:

S1. Twice the weight of Sohan is less than the weight of Mohan or that of Rohan.
S2. Twice the weight of Rohan is greater than the weight of Mohan or that of Sohan.

47. Which one of the following statements is correct?

(a) Weight of Mohan is greatest
(b) Weight of Sohan is greatest
(c) Weight of Rohan is greatest
(d) 'Whose weight is greatest' cannot be determined

Ans : (d) From statements S (I)
2S <M ⇒ S<M
2S <R ⇒ S<R
From State S (2)
2R > M ⇒ R⊁M
either R>M or R=M
and
2R > S ⇒ R⊁ S
either R > S or R = S
Hence,
Whose weight is greatest can not be determined.

48. Which one of the following statements is correct?

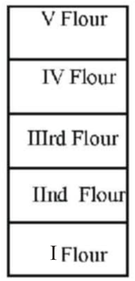
(a) Weight of Mohan is least
(b) Weight of Sohan is least
(c) Weight of Rohan is least
(d) 'Whose weight is least' cannot be determined

Ans : (b) From statement S(I)
2S < M ⇒ S < M
2S < R ⇒ S < R
From statement S(2)
2R > M ⇒ R ⊁ M
either R > M or R = M
and 2R > S ⇒ R ⊁ S
either R > S or R = S
Hence ,
weight of Sohan is least

49. A five-storeyed building with floors from I to V is painted using four different colours and only one colour is used to paint a floor.
Consider the following statements :
1. The middle three floors are painted in different colours.
2. The second (II) and the fourth (IV) floors are painted in different colours.
3. The first (I) and the fifth (V) floors are painted red.
To ensure that any two consecutive floors have different colour

(a) Only statement 2 is sufficient
(b) Only statement 3 is sufficient
(c) Statement 1 is not sufficient, but statement 1 along with statement 2 is sufficient
(d) Statement 3 is not sufficient, but statement 3 along with statement 2 is sufficient

Ans : (b)



Any two consectuve floors for different colours
Only statement 3 is sufficient
If the first and fifth floors are painted red, then the middle three floors will be of different colours. Because the total numbers of colours is 4.

50. In a conference, out of a total 100 participants, 70 are Indians. If 60 of the total participants are vegetarian, then which of the following statements is/are correct?
1. At least 30 Indian participants are vegetarian.
2. At least 10 Indian participants are non-vegetarian
Select the correct answer using the codes given below:

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2